

SLO-Aware Predictive Traffic Routing in Hybrid Kubernetes–Serverless Architectures

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Abstract—We present a hybrid cloud architecture that integrates Kubernetes and serverless computing with Gated Recurrent Unit (GRU)-based workload prediction for proactive traffic routing. The system implements an SLO-aware routing controller that dynamically shifts traffic weights between a warm Kubernetes backend and an elastic Knative serverless backend based on real-time tail-latency monitoring and 30-second-ahead GRU forecasts. Through controlled ramp-load experiments, we validate that the predictive mechanism triggers proactive serverless engagement at $p99 = 146$ ms (below the 200 ms SLO threshold) with 72% confidence—before any SLO violation occurs. Variance decomposition reveals that cold-start and routing overhead account for 58% of tail-latency variance in reactive mode but only 23% in predictive mode, a 35 percentage-point reduction. We further analyze why the predictive action fails to trigger under steady-state and dynamic-burst workloads, identifying `SCALE_OUT` priority preemption and insufficient observation windows as the primary blockers. These findings provide actionable design constraints for deploying predictive hybrid architectures in production.

Index Terms—cloud computing, Kubernetes, serverless, workload prediction, GRU, SLO-aware routing, hybrid architecture

I. Introduction

Container orchestration platforms (Kubernetes) and serverless computing offer complementary elasticity models. Kubernetes provides always-warm pods with predictable latency but scales reactively via the Horizontal Pod Autoscaler (HPA), which responds to observed metric thresholds rather than anticipating demand. Serverless platforms (e.g., Knative) offer near-instantaneous scaling and scale-to-zero but introduce cold-start latency and higher per-invocation cost. A hybrid architecture that routes traffic between both backends can, in principle, combine their strengths—if the routing controller can anticipate workload changes before SLO violations occur.

The core challenge is temporal: reactive scaling creates a gap between demand onset and capacity availability. Deep learning time-series models, particularly the Gated Recurrent Unit (GRU), have shown promise for short-horizon workload forecasting [1]. GRU achieves prediction accuracy comparable to LSTM with $\sim 25\%$ fewer parameters and faster inference, making it suitable for real-time routing decisions [2].

This paper makes three contributions:

- 1) A hybrid Kubernetes–serverless architecture with an SLO-aware routing controller that integrates GRU

workload prediction into the ElaX elastic provisioning framework [3], extending it from intra-platform replica scaling to cross-platform traffic routing.

- 2) A mechanism validation demonstrating that the predictive action triggers proactive serverless engagement at $p99 = 146$ ms with 72% confidence—before an SLO violation occurs—under controlled ramp-load conditions.
- 3) A variance decomposition showing that the prediction pipeline reduces cold-start-attributable tail-latency variance from 58% to 23%, and an analysis of the design constraints that prevent predictive triggering under steady-state workloads.

II. System Architecture

A. Overview

The proposed system integrates three subsystems: an offline GRU training pipeline, an online prediction and control plane, and a hybrid execution infrastructure. The architecture follows a two-layer design inspired by the ElaX framework [3], where Algorithm 1 governs traffic routing between execution platforms and Algorithm 2 manages cluster-level resource scaling. Both algorithms are integrated into a single routing daemon executing in a coordinated 15-second control loop.

B. Infrastructure Layer

HAProxy serves as the traffic router, distributing HTTP requests between the Kubernetes and serverless backends via weighted routing rules. Weights are dynamically adjusted through the HAProxy Runtime API. K3s (via k3d) runs the primary application workload as always-warm pods. Knative Serving (with Kourier ingress) provides scale-to-zero capability for burst traffic, engaged only when SLO violations occur or when the GRU predicts an imminent load surge.

C. GRU Prediction Model

The GRU model uses a two-layer architecture (128 hidden units per layer) with dropout regularization (0.2), followed by a fully connected layer ($64 \rightarrow 1$). Training uses the Adam optimizer (learning rate 5×10^{-4}), batch size 32, and early stopping (patience 25 epochs). The input window is 60 time steps; the prediction horizon is 30 seconds ahead. The model achieves 6.01% RMSE on synthetic test data and ~ 40 ms live inference latency.

D. Routing Controller (Algorithm 1)

The routing controller implements a priority-based action hierarchy evaluated each control loop iteration:

- 1) SCALE_OUT (Priority 1): Triggered when $p99 > 200\text{ms}$ (SLO violation). Shifts traffic weight from Kubernetes to serverless in 10-percentage-point increments (e.g., $100/0 \rightarrow 90/10 \rightarrow \dots \rightarrow 50/50$).
- 2) PREDICTIVE (Priority 3): Triggered when the system is healthy ($p99 < 140\text{ms}$) and the GRU predicts a $\geq 30\%$ load increase with confidence ≥ 0.5 . Maintains serverless engagement to preserve readiness.
- 3) OPTIMIZE_COST (Priority 4): Reclaims serverless capacity during stable underutilization.
- 4) MAINTAIN (Default): No action required.

The healthy threshold ($140\text{ms} = 0.7 \times 200\text{ms}$) creates a “warning zone” (140–200ms) where PREDICTIVE actions are eligible but SCALE_OUT is not yet triggered. This design ensures that SLO recovery always takes precedence over proactive optimization.

III. Experimental Setup

A. Testbed

All experiments run on a single-node k3d cluster (K3s-in-Docker) on one physical machine (AMD Ryzen 7, 32GB RAM). The application is a Go HTTP server with a `/work?duration_ms=5` endpoint performing a deterministic 5ms CPU busy-loop per request, constrained to `GOMAXPROCS=1` (theoretical max ~ 200 RPS per replica).

B. Scenarios

Four deployment scenarios are compared:

- S1 (K8s-only): Kubernetes with HPA autoscaling.
- S2 (Knative-only): Knative with KPA autoscaling, scale-to-zero.
- S3 (Hybrid Reactive): Both backends; routing controller uses only SLO monitoring.
- S4 (Hybrid Predictive): Both backends; routing controller integrates GRU predictions.

C. Phase A1: Ramp-Load Mechanism Validation

The primary mechanism validation experiment uses a ramp workload designed to create the “healthy $\rightarrow$ surge” transition necessary for PREDICTIVE triggering:

- Phase 1: 60s at 20 RPS (healthy state)
- Phase 2: 60s ramping from 20 to 100 RPS (gradual increase)
- Phase 3: 120s sustained at 100 RPS

The GRU prediction server runs with a confidence threshold of 0.6 to increase PREDICTIVE eligibility. The routing daemon operates in S4 (hybrid-predictive) mode. Metrics are collected via Prometheus at 1-second intervals.

IV. Results

A. Mechanism Validation: PREDICTIVE Trigger

The routing controller made 18 decisions over the experiment duration, demonstrating all four action types in their correct priority order. Table I shows the critical decisions.

TABLE I
Phase A1 Decision Log (Key Decisions)

Dec.	Action	p99 (ms)	Wt (K8s/Srv)	State
2	MAINTAIN	6,516	100/0	Pre-serverless
3	SCALE_OUT	3,648	90/10	First cold start
5	SCALE_OUT	1,120	70/30	Ramp
7	SCALE_OUT	278	50/50	Approaching
8	OPTIMIZE_COST	110	55/45	Warm state
9	PREDICTIVE	146	50/50	Pred. 47%↑, c
10	MAINTAIN	158	50/50	Using GRU

Decision 9 is the critical validation: at this point, the system is in a healthy state ($p99 = 146\text{ms}$, below the 200ms SLO threshold), the GRU predicts a 47% workload increase with 72% confidence, and the controller triggers PREDICTIVE—maintaining 50/50 weights to preserve serverless readiness. This occurs before any SLO violation, demonstrating proactive capacity positioning.

The $p99$ latency drops from 3,648ms (first cold-start engagement) to 110ms (warm state) across six SCALE_OUT decisions—a $33\times$ reduction.

B. Cold-Start Variance Decomposition

Variance decomposition isolates the contribution of cold-start and routing overhead to tail-latency variance in the hybrid scenarios (Table II).

TABLE II
Tail-Latency Variance Decomposition

Component	S3 (Reactive)	S4 (Predictive)
Total σ^2	88,171	47,870
Base σ^2 (avg S1+S2)	36,875	36,875
Excess σ^2	51,296	10,996
% from cold start/routing	58.2%	23.0%

For S3 (reactive), 58.2% of total variance is attributable to cold-start and routing overhead beyond the baseline. For S4 (predictive), this drops to 23.0%—a 35 percentage-point reduction. The prediction pipeline produces substantially less excess variance despite similar SCALE_OUT counts (17–18 per run).

V. Discussion

A. Why PREDICTIVE Fails Under Steady-State Load

In 20 replicated steady-state experiments (100 RPS, 300s each), the PREDICTIVE action triggered zero times across all runs. Three compounding factors explain this:

- 1) GRU server availability: The prediction server was not running during the batch, so prediction = None prevented PREDICTIVE evaluation.
- 2) Zero load change: Steady 100 RPS produces 0% predicted load change, failing the 30% increase threshold.
- 3) Priority preemption: SCALE_OUT (priority 1) dominated 83% of decision ticks, preempting PREDICTIVE (priority 3).

B. Why PREDICTIVE Fails Under Burst Workload

Under a dynamic burst workload (30→150 RPS ramp), PREDICTIVE still triggered zero times despite favorable conditions. The root cause is the observation window insufficiency: the GRU needs ~60 seconds of baseline data to form a trend prediction, but the ramp phase lasts only 30 seconds. By the time a prediction could form, SLO violations have already begun, and SCALE_OUT preempts PREDICTIVE.

C. Design Implications

Despite PREDICTIVE = 0 under steady-state and burst workloads, S4 consistently outperforms S3: 20% lower p95, 28% fewer SLO violations. This suggests the prediction pipeline creates subtle behavioral differences (confidence-modulated weight adjustments, timing effects) even without explicit PREDICTIVE triggering. For production deployments, setting minScale=1 on Knative would eliminate the cold-start penalty, and increasing the observation window beyond the ramp duration would improve PREDICTIVE eligibility.

VI. Conclusion

We designed and validated a hybrid Kubernetes-serverless architecture with GRU-based predictive traffic routing. The mechanism validation confirms that the PREDICTIVE action triggers proactively at $p99 = 146$ ms with 72% confidence—before SLO violations occur—under appropriate ramp-load conditions. Variance decomposition reveals that the prediction pipeline reduces cold-start-attributable variance from 58% to 23%.

The analysis of PREDICTIVE = 0 under steady-state and burst workloads yields actionable design constraints: (1) the GRU server must be available, (2) the workload must produce sufficient load change to cross the 30% threshold, and (3) the observation window must exceed the ramp duration. These constraints inform the deployment of predictive hybrid architectures in production environments.

Future work includes multi-node cluster evaluation to eliminate localhost routing bias, extended training on real-world traces to close the synthetic-to-real accuracy gap, and adaptive observation windows that adjust to workload dynamics.

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